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THIRD SEMESTER B.Tech

SUPPLEMENTARY END SEMESTER EXAMINATION FEB 2020

HU-201 ENGINEERING ECONOMICS

Time: 3:00 Hours Maximum Marks: 50

Note: Attempt any five questions.

All questions carry equal marks.

1. a) Discuss the difference between Microeconomics and Macroeconomics.

b) Following are the demand function for the commodities q_1 and q_2:

$$q_1 = p_1^{-1.3} p_2^{-0.7}$$

$$q_2 = p_1^{0.6} p_2^{-0.5}$$

Determine whether the commodities q_1 and q_2 are complementary or competitive.

2. a) Is inflation good for economy? How monetary and fiscal policy may be used to control inflation in India?

b) ABC Company is planning to expand its production operation. They identified three different technologies for meeting the goal. The initial investment and annual returns with respect to each of the technologies are summarised in table. Suggest the best technology which is to be implemented based on the present worth method of comparison assuming 20% interest rate, compounded annually.

	Initial outlay	Annual Returns	Life
Tech 1	12,00,000	4,00,000	10
Tech 2	20,00,000	6,00,000	10
Tech 3	18,00,000	5,00,000	10

3 . a)What do you mean by Green revolution and what are the drawbacks of the green revolution in Indian agriculture.

P.T.O.

b). Consider a monopolist who faces a linear demand function $p = 80-6q$ and also has a linear total cost function, $TC = 50+ 20q$. What will be the equilibrium level of output, price and profit? Also prove the second order condition for profit maximisation.

4. a) What is Balance of payment? Why would it be useful to examine a country's balance of payments data?

b) A company's management foresee that it will require Rs. 10, 00,000 at the end of 10th year for modernisation. How much it should deposit now to get the required amount after 10year. The rate of interest (r) is 9%.

5. a) How recent five- year plan of India is different from previous plans?

b) A company is planning to expand his business after ten years. To meet the expansion expenditure, at the end of first year, the company is planning to deposit Rs.10, 00,000 in the reserve and from the next year it will increase the amount to be deposited by Rs. 10,000 from the previous deposit for the next 9 years with an interest rate of 12%.Calculate the total amount which the company will have for expansion at the end of ten years.

6. a) Discuss the role of central Bank in liberalised economy.

b) Differentiate between:

i Tax and fees

ii Money Market and Capital Market

7. a) Why is firm under perfect competition a price taker and under monopolistic competition is price maker. Explain?

b) An almirah making firm has two options. Either it can use galvanised iron or aluminium as raw material. Both the materials provide equal services. Aluminium sheet costs Rs.90/m^2. And galvanised iron sheet costs Rs.70/m^2. Sheet of 60 m^2 size is required. The cost of finishing and bending of galvanised iron almirah is Rs.5000 while for aluminium, it is Rs.6050. if aluminium is used as raw material, it is joined by rivets, which costs Rs. 1250 per almirah. While for galvanised iron, it is Rs.1700 as spot welding is used. The transportation cost for each extra kilogram of weight is

Rs.850. an almirah made by galvanised iron weights 2 kg heavier than that made from aluminium. Find the better option and calculate economic advantage if all other costs incurred are same.

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